Synoptic Question 25: MHD Marine Propulsion

Reading Passage

Read the following article and then answer the questions that follow.

Magneto-Hydrodynamic (MHD) propulsion is a conceptual method for propelling marine vessels using electric and magnetic fields, without any moving parts like propellers. The basic principle involves passing seawater through a duct within the vessel. Within this duct, a strong magnetic field (B) is applied perpendicular to the direction of water flow, and an electric field (E) is applied perpendicular to both the flow direction and the magnetic field.

Seawater, being conductive due to dissolved salts, allows an electric current (I) to flow between electrodes placed on the sides of the duct. This current flows through the magnetic field, and the charged ions within the water experience a Lorentz force (F_L). According to Fleming's Left-Hand Rule, this force is directed along the duct, pushing the water out of the rear. By Newton's third law, this expulsion of water creates a forward thrust on the vessel.

The force density (f) on the fluid is given by the vector cross product $f = J \times B$, where J is the current density. For mutually perpendicular fields and flow, the magnitude of the total thrust F_{MHD} produced within the active volume V_{duct} of the duct is approximately $F_{MHD} = JBV_{duct}$. The current density J is related to the electric field E and the seawater conductivity σ by Ohm's Law for fluids, $J = \sigma E$.

A major challenge is generating the required powerful magnetic fields, often necessitating superconducting magnets. The vessel also experiences a hydrodynamic drag force F_D which typically increases with speed v, often modelled as $F_D = kv^2$ where k is a constant.

Data:

Data for a conceptual MHD-powered submersible:

Table 1: Data for MHD Submersible

Parameter	**Value**
Total mass of submersible, m	1.20×10^5 kg
Total volume of submersible hull, V_{hull}	118 m^3
Density of seawater, ρ_{sea}	1025 kg m^{-3}
Conductivity of seawater, σ	4.80 S m^{-1}
MHD duct length (active region), L	5.00 m
MHD duct width (electrode separation), w	0.80 m
MHD duct height, h	0.80 m
Magnetic field strength, B	4.00 T
Voltage applied across electrodes, V_{elec}	1600 V
Drag constant, k	1150 kg m^{-1}
Overall energy conversion efficiency (electrical to kinetic), η	45%

Questions

(a) The submersible is neutrally buoyant when fully submerged.

  (i) Calculate the buoyant force acting on the submerged submersible.

  (ii) Show that the mass of the submersible provided in the data is consistent with it being neutrally buoyant.

(b) The MHD drive generates thrust by applying electric and magnetic fields.

  (i) Calculate the electric field strength E between the electrodes.

  (ii) Calculate the current density J in the seawater within the duct.

  (iii) Calculate the total MHD thrust force, F_{MHD}, generated.

(c) The submersible starts from rest and accelerates.

(i) Calculate the initial acceleration of the submersible, assuming the MHD thrust is instantly at its full value.

(ii) Calculate the maximum theoretical speed (v_{max}) the submersible can reach when the drag force equals the MHD thrust.

(d) Consider the energy and power involved when the submersible travels at a constant speed of 8.00 m s^{-1}.

(i) Calculate the hydrodynamic drag force F_D at this speed.

(ii) Calculate the useful propulsive power P_{prop} being delivered to the water.

(iii) Given the overall efficiency η, calculate the total electrical power P_{elec} consumed by the MHD system.

(iv) Using $P = VI$, calculate the total current I drawn from the power source.

(e) The relationship between thrust F_{MHD} and applied voltage V_{elec} is being investigated.

(i) Show that the thrust is directly proportional to the applied voltage, $F_{MHD} \propto V_{elec}$.

(ii) The maximum speed v_{max} occurs when $F_{MHD} = F_D = kv_{max}^2$. Using this, derive a relationship between v_{max} and V_{elec}.

(iii) If the applied voltage were reduced by half, determine the new maximum speed the submersible could reach.

(f) Calculate the work done by the MHD drive against the drag force as the submersible travels a distance of 5.00 km at the constant speed of 8.00 m s^{-1}.

Solutions

(a) (i) Buoyant Force $F_B = \rho_{sea} V_{hull} g = (1025)(118)(9.81) = 1.185 \times 10^6$ N.

(ii) Weight $W = mg = (1.20 \times 10^5)(9.81) = 1.177 \times 10^6$ N.
Since $W \approx F_B$ (within reasonable rounding), the mass is consistent with neutral buoyancy.

(b) (i) $E = \frac{V_{elec}}{w} = \frac{1600}{0.80} = 2000$ V m^{-1}.

(ii) $J = \sigma E = (4.80)(2000) = 9600$ A m^{-2}.

(iii) Active Volume $V_{duct} = L \times w \times h = (5.00)(0.80)(0.80) = 3.20$ m^3.
$F_{MHD} = JBV_{duct} = (9600)(4.00)(3.20) = 122880$ N $\approx 1.23 \times 10^5$ N.

(c) (i) At rest, $v = 0$, so $F_D = 0$. Net force $F_{net} = F_{MHD}$.
$a = \frac{F_{net}}{m} = \frac{122880}{1.20 \times 10^5} = 1.024$ m s^{-2} ≈ 1.02 m s^{-2}.

(ii) Maximum speed occurs when $F_D = F_{MHD}$.
$kv_{max}^2 = F_{MHD}$
$v_{max} = \sqrt{\frac{F_{MHD}}{k}} = \sqrt{\frac{122880}{1150}} = 10.33$ m s^{-1} ≈ 10.3 m s^{-1}.

(d) Constant speed $v = 8.00$ m s^{-1}.

(i) $F_D = kv^2 = (1150)(8.00)^2 = 73600$ N.
(Note: This is less than F_{MHD}, so this speed is achievable).

(ii) At constant speed, Thrust = Drag. The useful force is equal to the drag force being overcome.
$P_{prop} = \text{Force} \times v = F_D \times v = (73600)(8.00) = 588800$ W ≈ 589 kW.

(iii) $\eta = \frac{P_{prop}}{P_{elec}} \implies P_{elec} = \frac{P_{prop}}{\eta}$
$P_{elec} = \frac{588800}{0.45} = 1.308 \times 10^6$ W ≈ 1.31 MW.

(iv) $I = \frac{P_{elec}}{V_{elec}} = \frac{1.308 \times 10^6}{1600} = 817.5$ A ≈ 818 A.
(Note: This calculation uses V_{elec} as the supply voltage, which is reasonable if the power source is matched to the electrode voltage).

(e) (i) $F_{MHD} = JBV_{duct} = (\sigma E)BV_{duct} = \sigma \left(\frac{V_{elec}}{w}\right) BV_{duct}$.
Since σ, w, B, V_{duct} are constants, $F_{MHD} = (\text{constant}) \times V_{elec}$.
Thus, $F_{MHD} \propto V_{elec}$.

(ii) From (i), $F_{MHD} = CV_{elec}$ where $C = \frac{\sigma B V_{duct}}{w}$.
At max speed, $F_{MHD} = kv_{max}^2$.
$CV_{elec} = kv_{max}^2 \implies v_{max}^2 = \frac{C}{k} V_{elec}$.
$v_{max} = \sqrt{\frac{C}{k}} \sqrt{V_{elec}}$.
Relationship: $v_{max} \propto \sqrt{V_{elec}}$.

(iii) Since $v_{max} \propto \sqrt{V_{elec}}$, if V_{elec} is halved, the new voltage $V'_{elec} = V_{elec}/2$.
$\frac{v'_{max}}{v_{max}} = \sqrt{\frac{V'_{elec}}{V_{elec}}} = \sqrt{\frac{V_{elec}/2}{V_{elec}}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}} \approx 0.707$.
New max speed $v'_{max} = v_{max} \times 0.707 = (10.33) \times 0.707 = 7.30$ m s^{-1}.

(f) Distance $d = 5.00$ km $= 5000$ m.
Drag force F_D at 8.00 m s^{-1} is 73600 N (from d(i)).
Work done $W = F_D \times d = (73600)(5000) = 3.68 \times 10^8$ J.

Cambridge International AS & A Level Physics (9702)

Paper 4: A Level Structured Questions **Variant 2 (Higher Difficulty)**

Topic focus: Magneto-hydrodynamic propulsion and linked physics concepts

Answer all questions. Show clear working, include units, and give answers to an appropriate number of significant figures.

Comprehension passage (read first)

A conceptual MHD submersible has a total mass of 1.25×10^5 kg and an external hull volume of 121 m^3. It operates in seawater of density 1025 $kg\,m^{-3}$ and conductivity 4.80 $S\,m^{-1}$. Inside the vessel, an MHD duct has active length 5.20 m, electrode separation 0.75 m, and height 0.90 m. A magnetic field of 4.50 T acts in the duct while a potential difference of 1800 V is applied across the electrodes. Resistive motion in water is modelled by $F_D = kv^2$, where $k = 1250$ $kg\,m^{-1}$. The overall efficiency from electrical input to useful kinetic output is $\eta = 0.42$. Use $g = 9.81$ $m\,s^{-2}$.

Extract the numerical values you need from the passage above for each question.

Data for Variant 2 (reference table)

Parameter	Value
Total mass of submersible, m	1.25×10^5 kg
External hull volume, V_{hull}	121 m^3
Density of seawater, ρ_{sea}	1025 $kg\,m^{-3}$
Seawater conductivity, σ	4.80 $S\,m^{-1}$
MHD duct active length, L	5.20 m
Electrode separation, w	0.75 m
MHD duct height, h	0.90 m
Magnetic flux density, B	4.50 T
Applied electrode voltage, V_{elec}	1800 V
Drag constant, k in $F_D = kv^2$	1250 $kg\,m^{-1}$
Overall conversion efficiency, η	0.42
Gravitational field strength, g	9.81 $m\,s^{-2}$

Questions (Variant 2 only)

(a) **Force balance and buoyancy** [8]

(i) Calculate the buoyant force acting on the fully submerged submersible. [2]

(ii) Calculate the weight of the submersible. [2]

(iii) Determine whether the submersible is positively buoyant, negatively buoyant, or neutrally buoyant. State the magnitude and direction of the resultant force. [2]

(iv) Suggest one practical engineering method to make the submersible neutrally buoyant at this depth. [2]

(b) **MHD thrust model** [11]

(i) Calculate the electric field strength between the electrodes. [2]

(ii) Calculate the current density in the seawater. [2]

(iii) Calculate the active duct volume. [1]

(iv) Using $F_{\text{MHD}} = JBV_{\text{duct}}$, calculate the thrust force. [2]

(v) Determine the initial acceleration when the submersible starts from rest. [2]

(vi) Calculate the maximum speed if drag follows $F_D = kv^2$. [2]

(c) **Power and efficiency at operating speed** [9]

The vessel cruises at a constant speed of $9.00\,\mathrm{m\,s^{-1}}$.

(i) Calculate the drag force. [2]

(ii) Calculate the useful propulsive power delivered to the water. [2]

(iii) Calculate the total electrical input power. [2]

(iv) Calculate the total current drawn from the power source using $P = VI$. [3]

(d) **Open-ended conceptual question: model limitations** [6]

The simple model uses uniform B, uniform current density, constant conductivity, and drag $F_D = kv^2$.

Discuss **three** physical effects that could cause the real maximum speed to be lower than predicted by the model. For each effect:

- identify the effect clearly,
- explain the physics,
- state how it reduces thrust and/or increases resistive losses.

(e) **Open-ended synoptic design question** [6]

A team proposes doubling the magnetic field strength while halving the electrode voltage, keeping all other dimensions unchanged.

Evaluate whether this is likely to improve high-speed performance and total energy efficiency. Your answer should include:

- a brief quantitative argument using proportional reasoning,
- at least one benefit,
- at least one drawback,
- a justified conclusion.

Sample Solutions (Variant 2)

(a) (i) $F_B = \rho V g = (1025)(121)(9.81) = 1.217 \times 10^6\,\mathrm{N}$.

(ii) $W = mg = (1.25 \times 10^5)(9.81) = 1.226 \times 10^6\,\mathrm{N}$.

(iii) $W > F_B$, so vessel is **negatively buoyant**. Resultant $= W - F_B = 9.1 \times 10^3\,\mathrm{N}$ downward.

(iv) Example method: adjust ballast tanks (admit a small volume of seawater or expel compressed air) until average density equals seawater density.

(b) (i) $E = V_{\mathrm{elec}}/w = 1800/0.75 = 2400\,\mathrm{V\,m^{-1}}$.

(ii) $J = \sigma E = (4.80)(2400) = 1.152 \times 10^4\,\mathrm{A\,m^{-2}}$.

(iii) $V_{\mathrm{duct}} = Lwh = (5.20)(0.75)(0.90) = 3.51\,\mathrm{m^3}$.

(iv)

$$F_{\mathrm{MHD}} = JBV_{\mathrm{duct}} = (1.152 \times 10^4)(4.50)(3.51) = 1.820 \times 10^5\,\mathrm{N}.$$

(v) At $v = 0$, drag is zero, so

$$a_0 = \frac{F_{\mathrm{MHD}}}{m} = \frac{1.820 \times 10^5}{1.25 \times 10^5} = 1.46\,\mathrm{m\,s^{-2}}.$$

(vi) At top speed, $F_{\mathrm{MHD}} = kv^2$:

$$v_{\max} = \sqrt{\frac{F_{\mathrm{MHD}}}{k}} = \sqrt{\frac{1.820 \times 10^5}{1250}} = 12.1\,\mathrm{m\,s^{-1}}.$$

(c) (i) $F_D = kv^2 = (1250)(9.00)^2 = 1.0125 \times 10^5\,\mathrm{N}$.

(ii) $P_{\mathrm{prop}} = F_D v = (1.0125 \times 10^5)(9.00) = 9.11 \times 10^5\,\mathrm{W}$.

(iii) $\eta = P_{\mathrm{prop}}/P_{\mathrm{elec}} \Rightarrow P_{\mathrm{elec}} = P_{\mathrm{prop}}/\eta$

$$P_{\mathrm{elec}} = \frac{9.11 \times 10^5}{0.42} = 2.17 \times 10^6\,\mathrm{W}.$$

(iv) $I = P/V = (2.17 \times 10^6)/1800 = 1.21 \times 10^3\,\mathrm{A}$.

(d) Possible points (any 3 developed): non-uniform magnetic field near duct edges reducing average JB force; heating/salinity changes altering σ and therefore lowering J for fixed voltage; turbulence and boundary-layer separation increasing effective drag above kv^2 model; bubble formation/electrolysis reducing conduction path; power electronics/internal resistance causing lower effective electrode voltage at high current.

(e) Doubling B and halving V gives $E \propto V$ so $J \propto V$ halves, but $F \propto JB$ so ideal thrust stays roughly unchanged. Potential benefit: lower voltage stress and insulation demands. Drawbacks: stronger magnets increase mass, cryogenic burden and parasitic power; additional system mass can increase displacement and drag. Therefore high-speed performance may not improve in practice; net efficiency likely worsens unless magnet system overhead is very small.

Mark Scheme (CIE P4 style) – Variant 2

Guidance: award marks for valid alternative methods. Penalise significant arithmetic errors once only where appropriate.

Q1 [8 marks]

- (a)(i) Uses $F_B = \rho V g$ and obtains 1.22×10^6 N (accept 1.21–1.22×10^6): M1 A1
- (a)(ii) Uses $W = mg$ and obtains 1.23×10^6 N (accept close values): M1 A1
- (a)(iii) Correct comparison and states negatively buoyant with downward resultant about 9×10^3 N: B1 B1
- (a)(iv) Valid practical method linked to buoyancy control (e.g. ballast mass/volume adjustment): B1 E1

Q2 [11 marks]

- (b)(i) $E = V/w$ with correct substitution and answer $2400\,\mathrm{V\,m^{-1}}$: M1 A1
- (b)(ii) $J = \sigma E$ and answer $1.152 \times 10^4\,\mathrm{A\,m^{-2}}$: M1 A1
- (b)(iii) $V_{\text{duct}} = Lwh = 3.51\,\mathrm{m^3}$: B1
- (b)(iv) $F = JBV$ and answer about 1.82×10^5 N: M1 A1
- (b)(v) $a = F/m$ at rest and answer about $1.46\,\mathrm{m\,s^{-2}}$: M1 A1
- (b)(vi) $v_{\max} = \sqrt{F/k}$ and answer about $12.1\,\mathrm{m\,s^{-1}}$: M1 A1

Q3 [9 marks]

- (c)(i) $F_D = kv^2$ and answer 1.0125×10^5 N: M1 A1
- (c)(ii) $P_{\text{prop}} = Fv$ and answer 9.11×10^5 W: M1 A1
- (c)(iii) Uses $\eta = P_{\text{prop}}/P_{\text{elec}}$ and answer 2.17×10^6 W: M1 A1
- (c)(iv) Uses $I = P/V$ with consistent values and answer 1.21×10^3 A: M1 A1 B1

Q4 [6 marks]

- Award up to 2 marks per developed effect (1 identification + 1 physics explanation) for any three relevant effects.
- Typical valid effects: non-uniform field/current, conductivity changes with temperature/salinity, turbulence/wake losses, electrolysis/gas bubble effects, internal electrical losses.
- Cap at 6 marks.

Q5 [6 marks]

- Correct proportional argument that ideal thrust remains approximately unchanged when $B \times V$ is unchanged: B2
- One relevant advantage explained: B1
- One relevant disadvantage explained: B1
- Balanced, justified conclusion on performance and efficiency: E2

Total: 40 marks